

Sparsh Chakraborty

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EDUCATION

Bachelors of Computer Science Engineering with specialization in AI & ML,

[Sep'22 - May'26]

3rd year student of the School of Computer Science at Vellore Institute of Technology Chennai Campus, expected graduation in May 2026. CGPA: 8.97/10

- **Coursework:** Data Structures and Algorithms, Algorithm Analysis and Design, Object Oriented Programming, Computer Networks, Operating Systems, Database Systems, Machine Learning

TECHNICAL PROJECTS

MRIG [\[Link\]](#)

- Developed medical report generation software utilizing Resnet50 CNN model to extract 13 parameters from chest X-ray scans. Led frontend and backend integration using Flask, HTML/CSS, and Vanilla JS. Successfully deployed ML model and integrated ChatBot with custom dataset and LangChain.

VitaFile [\[Link\]](#)

- Spearheaded VitaFile's development, leveraging Google's APIs for centralized EHR access and pioneering multilingual support, voice interaction, and gesture recognition for intuitive user experience.

Jalamniti [\[Link\]](#)

- Developed the front-end of a machine learning model-based web application that displayed the contamination levels in nearby water sources.
- Utilized Google technologies to enhance the functionality and user experience of the web application.
- Developed a ML based Website to monitor the Contamination levels and pollution levels in nearby Water Bodies.

EXPERIENCE

Software Development Intern, Stair Digital Pvt Ltd

[Sep'23-Feb'24]

- Developed responsive web applications using React, Django, and PostgreSQL, contributing to key features and enhancing user engagement.
- Automated data extraction processes through web scraping, enabling efficient data analysis and support for strategic decision-making.
- Implemented AI models for predictive analysis, improving operational efficiency by 20% through machine learning optimizations.

AI and Robotics Intern, StemTec Pvt Ltd (VIT Chennai Incubated)

[Feb'24-Present]

- Pivotal in the development of Autonomous Ground Vehicle (AGV), employing cutting-edge Computer Vision techniques with RealSense SDK and PyTorch Transformers.
- Directed and motivated a team of 5, instilling a collaborative ethos and dynamic leadership within the workplace environment.
- Seamlessly integrated computer vision models into our prototype via ROS2, ensuring cohesive functionality and optimal performance.

AI Research Intern, Samsung Prism

[Apr'24-Present]

- Led Development of Innovative Audio Restoration System using CNNs, LSTMs, and Autoencoders for enhanced audio clarity.
- Integrated Advanced Deep Learning Models for precise audio degradation classification and targeted spectral-temporal restoration.
- Optimized System for Real-Time Processing, implementing model pruning and quantization for efficient deployment on limited-resource devices such as the new line Galaxy Phones.

PUBLICATIONS

- Deep Neural Network-based Speech Steganography with Embedded Images
National Conference on Advancements in Computer Applications (NCACA'24)
[\[Certificate\]](#)
[\[Paper\]](#)

POSITIONS OF RESPONSIBILITY

Google Developer Student Club, VIT Chennai | Social Media Lead [Aug'23 - Present]

- Member of the Web Development team but leading a team for our Social Media Outreach
- Conducted events for Google DSC and grew our Social Media Presence
- Recruited Members for the community and helped them grow

IEEE Student Branch | Content lead [Aug'23-Present]

- I oversee the creation and publication of content related to prompt engineering and AI for the IEEE SB.
- I provide guidance and direction to my team members to ensure that the content produced is accurate, informative, and engaging.
- I am responsible for the development and maintenance of the CMS used by my team to manage and publish content on the IEEE SB website.

SKILLS

Languages and Framework: C, C++, Python3, Solidity Java, HTML5, CSS3, JavaScript, ReactJS, NodeJS, SQL, Django, Vue.js, FastApi, ROS2

Software: Git, MATLAB, Keil UVision, LTspice

Spoken Languages: English (Full Professional Proficiency), Hindi (Native Proficiency), Bengali (Native Proficiency)